Time-Series SAR Monitoring of Rice in Multiple Cropping Modes Combining Statistical and Phenological Characteristics

Lihua Wang, Tong Wang, Hao Ma, Peng Lu, Weiwei Sun, *Senior Member, IEEE*, Li Fan, Hongmei Wang, Yuchen Wu, and Yumiao Wang

***Abstract*— Rice is a critical staple crop worldwide. Accurately monitoring rice cultivation areas plays a vital role in supporting the governmental formulation of food policies and agricultural sustainability. Synthetic aperture radar (SAR) images are not affected by weather and can acquire backscatter information below the rice canopy, which is beneficial for long-term continuous monitoring of rice. However, most existing studies made less consideration to integrated research on rice cultivation with diverse planting modes under complex terrain, which makes it difficult to reflect the real situation of rice cultivation. Therefore, in Ningbo, China, where the rice planting modes have single-crop early, middle, late rice, and double-crop rice, the rice fields are fragmented, and the terrain is complex, which is selected as the experimental area. Four types of rice features (SAR backscatter, topographic, statistical, and phenological features) in different cropping modes were thoroughly explored, and the feature datasets were correspondingly constructed using all available Sentinel-1 SAR images in 2022. The combination of statistical and phenological characteristics resulted in the highest recognition accuracies for single-crop early, middle, late rice, and double-crop rice, with F1-scores of 0.85, 0.84, 0.89, and 0.89, respectively. In addition, the reliability of the multiple cropping mode rice identification framework in this study is discussed quantitatively and qualitatively in terms of the temporal resolution of the time-series images, the smoothing method of the time-series SAR data, and the interference of other land cover types. This study can provide a reference for other rice planting areas with complex rice cropping modes.**

***Index Terms*— Multiple cropping modes, paddy rice, phenological characteristics, statistical characteristics, time-series Sentinel-1.**

I. Introduction

PADDY rice plays a vital role in maintaining global food security, particularly in Asian countries such as China, where it serves as a fundamental dietary component for over half of the populace [1], [2], [3], [4]. As a semi-aquatic crop, the cultivation of paddy rice requires the presence of standing water throughout its growth stages, which contributes to the considerable consumption of water resources [5], [6], [7]. Moreover, paddy rice fields are a significant source of carbon emissions in agriculture. Therefore, monitoring the area of paddy rice cultivation is crucial for studying climate change and its impact [8], [9]. However, various factors, such as climate change, water scarcity, degradation of soil fertility, changes in cropping systems, and labor migration, have led to a decline or slower growth in rice acreage and production levels across most nations in recent years [10], [11]. It is necessary to address these challenges and find sustainable solutions to ensure the stability and productivity of paddy rice cultivation.

Compared to traditional field survey methods, the use of remote sensing technology allows for the large-scale and long-term monitoring of paddy rice, providing better economic benefits [6], [12], [13], [14]. The two main sources of remote sensing data utilized for paddy rice mapping are optical remote sensing images and synthetic aperture radar (SAR) images. The values of image pixels within optical remote sensing images show a correlation with vegetation cover, enabling a more precise assessment of crop growth status [13], [14]. The optical remote sensing data comprise the red, green, blue, red-edge, and near-infrared bands, which are highly responsive to vegetation information [15], [16], [17], [18]. This makes them particularly well suited for generating paddy rice maps on a large scale. However, the effectiveness of optical remote sensing imaging is heavily influenced by varying weather conditions. Specifically, the paddy rice planting areas in the southern regions of China often experience cloudy and foggy conditions, resulting in limited availability of high-quality optical remote sensing data. Consequently, accurately studying

Manuscript received 22 May 2024; revised 18 June 2024 and 30 June 2024; accepted 9 July 2024. Date of publication 18 July 2024; date of current version 15 August 2024. This work was supported in part by the Public Projects of Ningbo City under Grant 2023S102, in part by Zhejiang Provincial Natural Science Foundation of China under Grant LY22D010002, in part by the Key Projects of Major Scientific and Technological Tasks in Ningbo under Grant 2021Z107, in part by the Key Technology Breakthrough Plan Project of Science and Innovation Yongjiang 2035 under Grant 2024Z262, in part by Chongqing Natural Science Foundation under Grant CSTB2022NSCQ-MSX0588, in part by the National Natural Science Foundation of China under Grant 42122009, in part by Zhejiang Province "Pioneering Soldier" and "Leading Goose" R&D Project under Grant 2023C01027, in part by the Ningbo Science and Technology Innovation 2025 Major Special Project under Grant 2021Z107 and Grant 2022Z032, and in part by the Youth Scientist Project National Key R&D Program of China under Grant 2023YFF1305600. *(Corresponding author: Weiwei Sun.)*

Lihua Wang is with the Department of Geography and Spatial Information Techniques and Donghai Institute, Ningbo University, Ningbo, Zhejiang 315211, China (e-mail: wanglihua1@nbu.edu.cn).

Tong Wang, Weiwei Sun, Hongmei Wang, Yuchen Wu, and Yumiao Wang are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: 2011073004@nbu.edu.cn; sunweiwei@nbu.edu.cn; 1298140836@qq.com; 2014856003@qq.com; wanyumiao@whu.edu.cn).

Hao Ma is with the State Key Laboratory of Estuarine and Coastal Research, East China Normal University, Shanghai 200241, China (e-mail: mahao629@163.com).

Peng Lu is with the College of Geoexploration Science and Technology, Jilin University, Changchun 130026, China (e-mail: lupeng@jlu.edu.cn).

Li Fan is with Chongqing Institute of Meteorological Sciences, Chongqing 401147, China (e-mail: fanli_0223@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3430528

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

rice cultivation becomes a challenging task. Moreover, the interaction between optical remote sensing and the rice leaf canopy mainly involves absorption and reflection processes, and the leaf canopy often obscures information regarding the underlying rice structure and area, further complicating the retrieval of such details from optical remote sensing images [19].

SAR utilizes microwaves that remain unaffected by inclement weather conditions and darkness, enabling it to penetrate the dense rice canopy and gather data both on and beneath it. It offers the advantage of multipolarization and multimode observations [20], [21], [22]. Therefore, it is more suitable for paddy rice mapping in areas with cloudy weather and is an attractive alternative to optical remote sensing. Previous studies have attempted to determine the optimal timing for rice identification by combining rice phenology with a single-temporal SAR image [23], [24]. However, determining the optimal time in large study areas using just one SAR image proves challenging due to variations in rice phenology based on factors such as climate, water source, and farmers' subjective decisions. To overcome this limitation and achieve higher accuracy, the utilization of time-series SAR images has proven successful because time-series SAR images enable the observation of the entire phenological cycle of rice, capturing growth status and phenological changes, and facilitating the extraction of more discriminative rice features [19], [25]. Researchers have already explored time-series SAR-based rice identification in various areas [20], [26], [27], [28] mainly concentrating on the plain regions with extensive rice cultivation and relatively simplified cropping modes (such as single or double crop) found in the Northeast Plain of China [29], [30], the Sichuan Basin [31], [32], [33], and the Mississippi River Basin of the United States [34]. However, there is still insufficient research conducted in complex terrain regions with multiple cropping modes, large variations in rice patch scales, and fragmented patches.

For the characteristics used to identify rice fields in complex terrain regions, previous studies have mainly focused on the differences in backscattering coefficients between rice and nonrice at different phenological periods [28], [35], [36], [37], [38] and these differences are useful for distinguishing between rice and nonrice field types. However, rice cultivation comprises various planting modes (e.g., single-crop early rice, single-crop meddle rice, single-crop late rice, and double-crop rice), and rice of different planting modes often coexists in the same region simultaneously. Previous studies seldom considered the differences in the time and intensity of backscattering coefficient changes, the differences in phenology, and the differences in topographic indexes of different rice fields in areas with multiple cropping modes and thus could not accurately recognize different rice fields in areas with multiple cropping modes (e.g., single-crop early rice, single-crop meddle rice, single-crop late rice, and double-crop rice), resulting in a limited view of rice cultivation within a region.

In this study, the variability of backscattering coefficients, their statistical characteristics, phenological parameters, and topographic features of different rice types are considered comprehensively to achieve finer rice identification in areas with multiple cropping modes. The main objectives include the following:

1) characterizing the distinguishing features of rice in multiple cropping modes and constructing a corresponding rice feature dataset for each specific cropping mode;
2) assessing the contribution of different features to the identification of single-crop early, middle, late rice, and double-crop rice;
3) achieving high-precision rice recognition and conducting a qualitative and quantitative assessment of the sensitivity of rice identification methods.

II. Case Study Area and Data

A. Case Study Area

Ningbo, in Zhejiang Province, China, is a mixed cropping area, where single-crop early, middle, late rice, and double-crop rice are typically grown. The smaller size of rice paddies in Ningbo is attributed to the topography, with the terrain being higher in the southwest and lower in the northeast. The region consists of various topography types, including mountains, hills, basins, and plains, with altitudes ranging from 0 to 1018 m (Fig. 1). It encompasses ten county-level cities, counties, and districts, covering a total land area of 9816 km^2 (Fig. 1). The climate in this area is classified as subtropical to tropical. The average annual temperature is 16.4 °C, and the average annual precipitation is 1480 mm.

The crop structure and rice cultivation system in Ningbo are complex and diverse. Rice, wheat, potatoes, and sweet potatoes are the main types of food crops, accounting for 68%, 16%, 10%, and 6% of the total cultivated area, respectively [39]. The growing period for rice in Ningbo typically ranges from four to six months (Fig. 2). There are two cropping systems: single-crop rice (early, middle, or late rice) and double-crop rice (early and late rice). The transplantation of middle rice typically occurs from mid- to late-June followed by its harvest between the months of September and October. As for early rice, it is commonly transplanted during late April and then harvested in early July. On the other hand, late rice is cultivated on the same field as early rice and is transplanted from mid- to late-July, with the harvesting period spanning from late September to October.

B. Sentinel-1 SAR Data and Preprocessing

We utilized all available Sentinel-1A interferometric wide (IW) mode high-resolution Level-1 ground range detected (GRD) products that covered Ningbo City in 2022. Unfortunately, Sentinel-1B data for 2022 were not accessible due to a system failure on December 23, 2021. The spatial resolution was 20 m in the range direction and 22 m in the azimuth direction, with a pixel size of 10 × 10 m. The temporal resolution was 12 days. The data consisted of two polarization bands: VH and VV.

Preprocessing of Sentinel-1A data involves multiple steps, including thermal noise removal, radiometric calibration, terrain correction, and conversion of terrain-corrected values to decibels (backscatters to dB) via logarithmic scaling. For terrain correction, the shuttle radar topography mission (SRTM)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

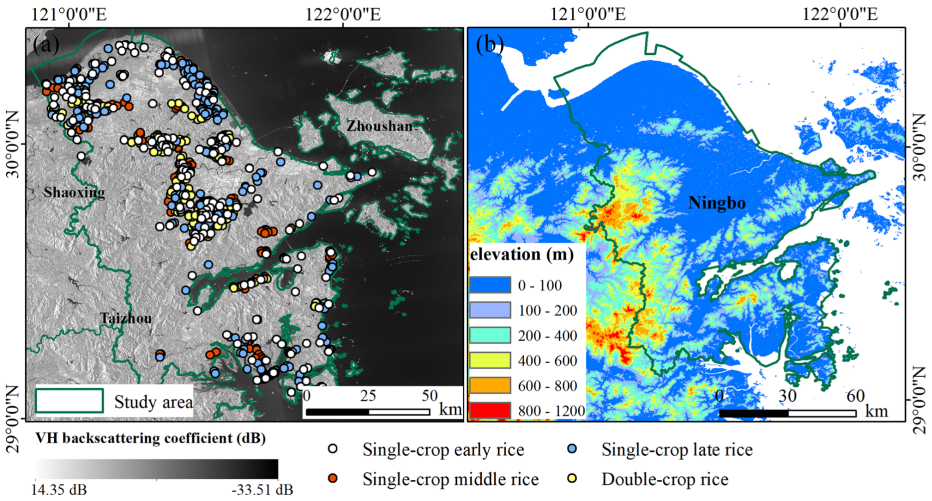


Fig. 1. (a) Location and samples. (b) Topography of the study area.

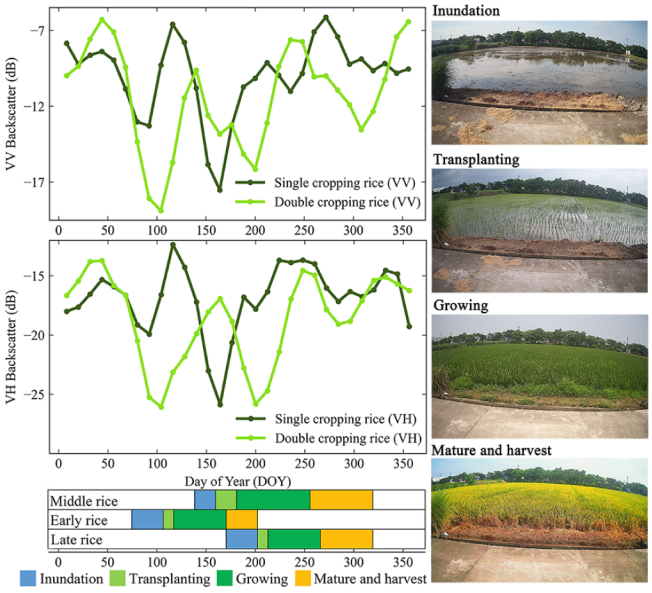


Fig. 2. Paddy rice planting calendar, Sentinel-1 VV and VH time series during the paddy rice growth period, and field photographs of rice.

data are used. The resulting terrain-corrected values are then converted to decibels using a logarithmic scaling method, as shown in the following equation:

$$\delta = 10 * \log_{10} \delta_0 \tag{1}$$

where δ_0 is the original backscatter coefficient and δ is the backscatter coefficient after backscatter to dB.

In addition, SAR backscatter intensity can still be influenced by various environmental conditions, including short-term heavy rainfall and the diverse paddy rice calendars implemented by farmers [19], [22]. These conditions could result in significant changes in backscatter intensity concerning rice phenology. Therefore, we applied a Savitzky–Golay filter with a three-fit polynomial and a three-length window to smooth the time-series Sentinel-1 data [40], [41].

C. ALOS DEM

ALOS DEM data refer to elevation data collected by the advanced land observing satellite (ALOS) satellite phased-array L-band SAR (PALSAR). The accuracy of ALOS DEM data is up to 12.5 m both horizontally and vertically. We obtained the ALOS DEM data from Alaska Satellite Facility (ASF) (https://search.asf.alaska.edu) and conducted mosaicking and resampling to ensure a consistent pixel spacing of 10 × 10 m, matching the Sentinel-1 IW GRD data.

D. Auxiliary Data

The true value sample data were obtained from rice field photographs, UAV images, and Google Earth images. From July to October 2022, we collected UAV multispectral images of local farmland areas in Ningbo and obtained a large number of rice field photographs with time and location information. Finally, a total of 1600 samples (training samples: 800 and validation samples: 800) were obtained, including 200 (100/100) single-crop early rice samples, 200 (100/100) single-crop middle rice samples, 200 (100/100) single-crop late rice samples, 200 (100/100) double-crop rice samples, and 800 (400/400) nonrice samples. Then, we checked each sample one by one to ensure its reliability and typicality. In addition, we also acquired real samples of single- and double-crop rice fields from the UAV multispectral images of the farmland area to verify the accuracy of rice field mapping in a more detailed way.

III. Methodology

In this article, we proposed a framework for integrated monitoring of rice in multiple planting modes combining statistical and phenological characteristics based on time-series SAR, as shown in Fig. 3. We first preprocessed the Sentinel-1 SAR data and ALOS DEM data. Then, time-series SAR backscattering characteristics were analyzed for different cropping modes of rice (single-crop early, middle, late rice, and double-crop rice). Based on this, four types of features that were generated for different cropping modes of rice,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

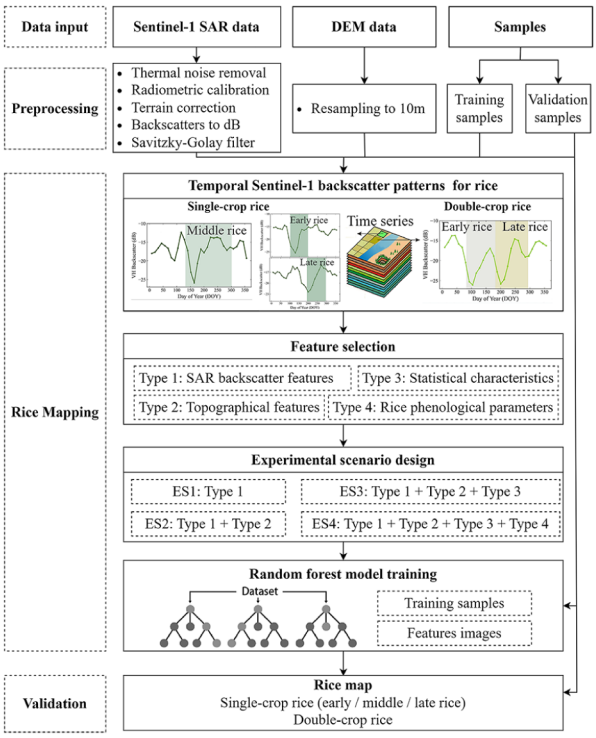


Fig. 3. Framework for integrated monitoring of multiple rice planting modes combining statistical and phenological characteristics based on time-series SAR.

including SAR backscatter coefficients, topographic features, statistical characteristics, and rice phenological parameters, were extracted. Furthermore, to evaluate the contribution of the four types of features in identifying single-crop early, middle, late rice, and double-crop rice, we designed four experimental scenarios. Subsequently, a random forest classifier was employed to implement rice identification and create maps for single-crop early, middle, late rice, and double-crop rice. Finally, we assessed the accuracy of the rice maps by using validation samples and rice area statistics.

A. Temporal Sentinel-1 Backscatter Modes for Different Land Cover Types

The Sentinel-1 VV and VH time-series curves for different land cover types were generated using the training samples (Fig. 4). Since SAR signals are sensitive to the structure of surface objects and different surface objects have different backscattering mechanisms, there are differences in the SAR backscatter coefficients and temporal variation pattern for different land cover types [38].

1) Single-Crop Rice: Single-crop rice comprises three types: single-crop early rice, middle rice, and late rice. The transplantation process for single-crop early rice usually occurs around mid-April (DOY100). At this particular time, the rice field exhibits abundant water content and a smooth surface, resulting in a VH backscattering coefficient of −25 dB or below for single-crop rice. As the rice plants grow and develop, their canopy cover increases, replacing the water body as the primary cover in the paddy field. The VH and VV backscattering coefficients of single-crop early rice gradually rise due to the amplified secondary scattering effects, eventually reaching their peak values of −15 dB in late June (DOY170). Subsequently, these backscattering coefficients gradually exhibit a significant declining trend once the single-crop early rice reaches maturity. After the rice harvest, nonrice crops such as oilseed rape are typically cultivated in the rice fields.

With regard to the single-crop middle rice, the alterations in the backscattering coefficient closely resembled those observed in the single-crop early rice; however, there were discrepancies in terms of phenological dates. The single-crop middle rice was transplanted between the conclusion of May and the conclusion of June (DOY 140–180). Correspondingly, the backscattering coefficient remained low during this period, exhibiting a gradual increase from the conclusion of June to the middle of September (DOY 180–250) when the rice was undergoing growth and maturation. Following the harvest of the middle rice, various other crops were subsequently cultivated within the same field.

As for the single-crop late rice, it is planted in the middle-to-early stages of July (DOY 190) subsequent to the harvesting of wheat or oilseed rape. Upon transplantation, the backscatter coefficient experienced a monotonic increase from −25 to −15 dB as the rice plants progressively developed. It matures for harvest by late September or even October.

2) Double-Crop Rice: Double-crop rice is comprised of two annual cycles: early rice and late rice. This results in distinct backscattering coefficient patterns. In mid-April, the transplantation of early rice takes place, where the main backscattering mechanism is specular reflection. As a result, the backscattering coefficient is relatively low, measuring around −25 dB. Subsequently, from late April to the end of June, the early rice enters a growth period, leading to a gradual increase in the backscattering coefficient from −25 to −15 dB. As the rice reaches maturity at the end of June, its backscattering coefficient starts to decline. After harvesting the early rice in early July, the soil is plowed, and irrigation begins to prepare for transplanting late rice. By mid-July, the late rice is planted. Once the rice is transplanted, it enters its growing period, and the backscattering coefficient undergoes its second cyclic change of the year. In late September or even October, the late rice is fully grown and harvested, after which the paddy field enters the fallow stage.

3) Nonrice Land Cover Types: For build-up areas, double scattering between the ground and walls mainly results in high backscattering intensity, with VV and VH backscattering coefficients fluctuating around −3 and −8 dB, respectively. On the contrary, the backscattering intensity of the water body is low because the SAR signal mainly undergoes specular reflection after reaching the water surface. This very high or low backscattering intensity results in buildings and permanent water bodies hardly interfering with the identification of rice. For natural vegetation, SAR signals produce body scattering in the canopy, resulting in higher backscattering intensities with VV and VH backscattering coefficients ranging from −9.33 to −3.89 dB and −15.58 to −10.63 dB, respectively. The backscatter intensity of dry crops is lower than that of natural vegetation, and its dynamics are mainly related to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

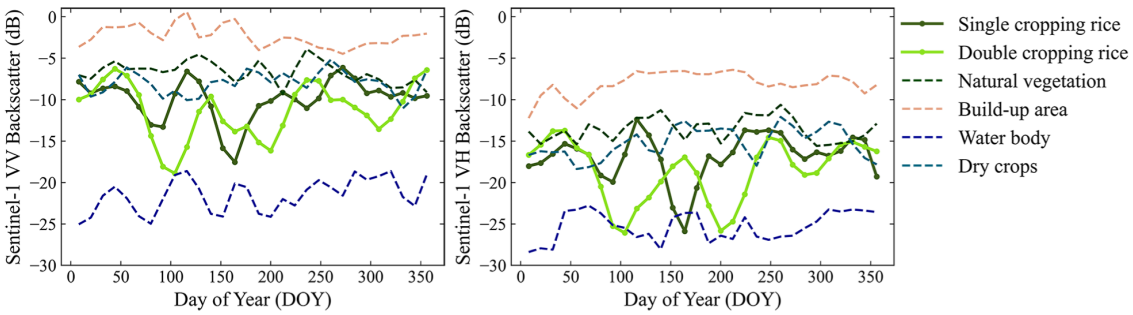


Fig. 4. Examples of Sentinel-1 VV and VH time series of different land cover types.

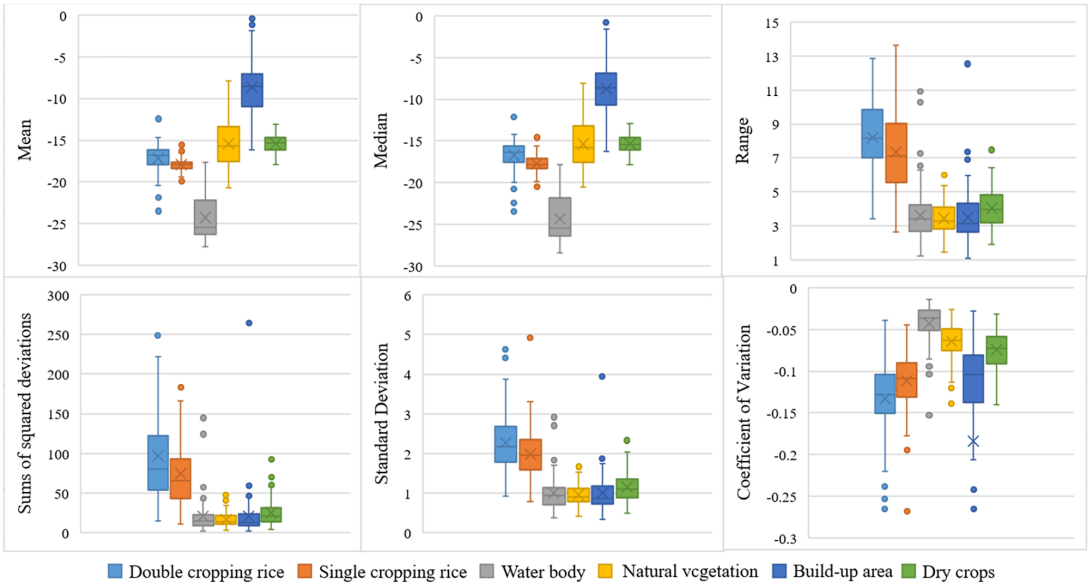


Fig. 5. Differences in statistical characteristics of rice and nonrice land cover types based on VH backscatter coefficients.

the growth state of the crop and the prevailing phenological stage.

B. Feature Selection

1) Sentinel-1 Backscatter Coefficients: Based on the analysis of temporal Sentinel-1 backscatter modes for different land cover types in Section III-A, we clarified that the SAR backscatter coefficients and their modes of variation with phenology for rice are different from those of nonrice land cover types, which is the basis for SAR-based identification of rice.

In order to recognize rice with high accuracy in areas with multiple cropping modes, we extracted statistical characteristics and rice phenology parameters using the time-series SAR backscatter coefficients.

2) Statistical Characteristics: The statistical characteristics include the mean, median, range, sum of squared deviations, standard deviation, and coefficient of variation of the SAR backscatter coefficients in a year. These characteristics provide insights into the general level, degree of dispersion, and distribution of backscatter coefficients for rice throughout the year. Fig. 5 illustrates the variations in statistical characteristics among single-crop rice, double-crop rice, and other nonrice land cover types. Specifically, the mean and median values of backward scattering coefficients for single- and double-crop rice were mainly located between −20 and −15 dB, which were significantly higher than those for water bodies and lower than those for built-up areas. However, the mean and median values could not distinguish rice (single-crop rice and double-crop rice) from natural vegetation and dry crops more accurately. In response to this, statistical characteristics reflecting the degree of dispersion of the backscattering coefficients (range, sum of squared deviations, and standard deviation) can distinguish them. Compared with dry crops and natural vegetation, single-crop rice and double-crop rice have a greater degree of dispersion, and their ranges, sum of squared deviations, and standard deviations are greater than 5, 40, and 1.5, respectively. In addition, the coefficient of variation can also differentiate rice (single-crop rice and double-crop rice) from waterbodies and natural vegetation to a certain extent. Therefore, statistical characteristics can well distinguish rice (single-crop rice and double-crop rice) from nonrice land cover types, but not single-crop rice and double-crop rice.

3) Phenological Parameters: To address the challenge of distinguishing between single-crop rice and double-crop rice using statistical characteristics, we focused on extracting phenological parameters. These parameters were identified as effective indicators for distinguishing between single-crop early, middle, late rice, and double-crop rice. The analysis of the temporal Sentinel-1 backscatter modes of single-crop rice

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DEFINITION AND DESCRIPTION OF RICE PHENOLOGICAL PARAMETERS

Num.	Rice phenological parameters	Description
1	Rice transplanting date (TD)	The day of year (DOY) of the σ0 reaches the local annual minima
2	Rice agronomy flooding decline speed (Speed (R_{ST}))	The ratio of the σ_0 change over the number of days from paddy rice field submerged date (SD) to TD
3	Rice mature date (MD)	DOY when the σ_0 reaches to the maximum
4	Green-up speed (Speed (R_{TM}))	The ratio of the σ_0 change over the number of days from paddy rice TD to MD
5	Growing season length (GSL)	The length from paddy rice TD to MD

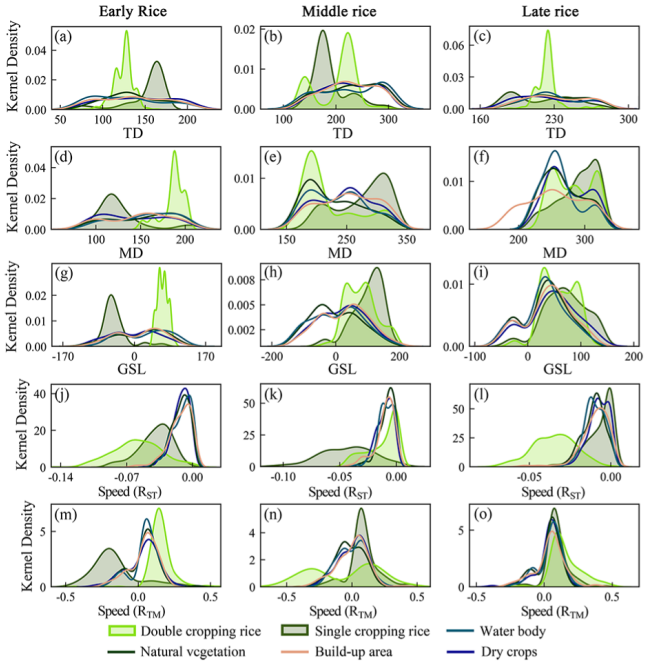


Fig. 6. Differences in phenological parameters of rice and nonrice land cover types based on VH backscatter coefficients. The first, second, and third columns respectively display histograms of various land cover types on early, middle, and late rice phenological parameter images (Table I), illustrating the role of phenological parameters in rice identification tasks. (a)–(o) Plotted based on sample points of various land cover types and phenological parameter images for early, middle, and late rice.

and double-crop rice in Section III-A served as the basis for this extraction (Table I). The differences in the phenological parameters of early, mid, and late rice between single-crop rice and double-crop rice are visually presented in Fig. 6. In Fig. 6, we extracted five phenological parameters (Table I) based on the time-series Sentinel-1 VH images over the early, middle, and late rice phenology periods, for a total of 15 phenological parameters, which are actually 15-scene feature images. Fig. 6 was then plotted based on these feature images and sample points to show which features are more useful in distinguishing different rice field types.

4) Topographical Features: Altitude affects temperature, thus limiting the extent to which rice can be grown.

TABLE II
FOUR EXPERIMENTAL SCENARIOS FOR MAPPING RICE AND ES REPRESENTS EXPERIMENTAL SCENARIOS

	ES1	ES2	ES3	ES4
SAR backscatter coefficients	√	√	√	√
Topographic features		√	√	√
Statistical characteristics			√	√
Rice phenological parameters				√

In addition, rice growth requires irrigation, so rice cultivation requires flatter microtopography. Therefore, elevation and slope factors are also considered.

C. Experimental Scenarios and Random Forest Classifier

To evaluate the contribution of four types of features to map single-crop early, middle, late rice, and double-crop rice, we designed four experimental scenarios (Table II). ES1 used only SAR backscatter coefficients to extract rice. ES2 used SAR backscatter coefficients and topographic features to explore the role of elevation and slope. ES3 added statistical characteristics to ES2 to quantitatively assess the contribution of statistical characteristics in rice identification. ES4 added rice phenological parameters to ES3 to quantitatively assess the contribution of rice phenological parameters in rice identification.

Random forest is an integrated algorithm that uses multiple decision trees and sample data to train a model and implement predictions. The method can run on large datasets, evaluate the importance of each feature, and has excellent accuracy, good parallelism, and scalability. In this study, the rice identification in this article is implemented by the "ee.Classifier.smileRandomForest" tool in the Google Earth Engine (GEE) platform. In order to balance the computational efficiency of the algorithm and the accuracy of the results, we tried different numbers of trees (20, 50, 100, and 150) in the GEE platform and finally chose 100 trees because they achieved relatively high accuracy and efficiency; all other parameters of the algorithm were set by default.

IV. RESULTS

A. Performance of Different Scenarios

The accuracy metrics (Table III) for single-crop early rice, middle rice, late rice, and double-crop rice were found to be at their minimum values when using only SAR backscatter coefficients (ES1). This suggests that SAR backscatter coefficients alone are not sufficient for accurately recognizing rice in multiple cropping modes. Compared with ES1, ES2 with the incorporation of topographic features improved the recognition accuracies of single-crop early rice, middle rice, late rice, and double-crop rice, especially the $F1$-score of single-crop late rice improved by 0.07 (from 0.82 to 0.89). Compared to ES2, ES3 with the incorporation of statistical characteristics improved the $F1$-score of single-crop early rice and middle rice by 0.04. Overall, we found that the $F1$-score of single-crop early, middle, late rice, and double-crop rice when four

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

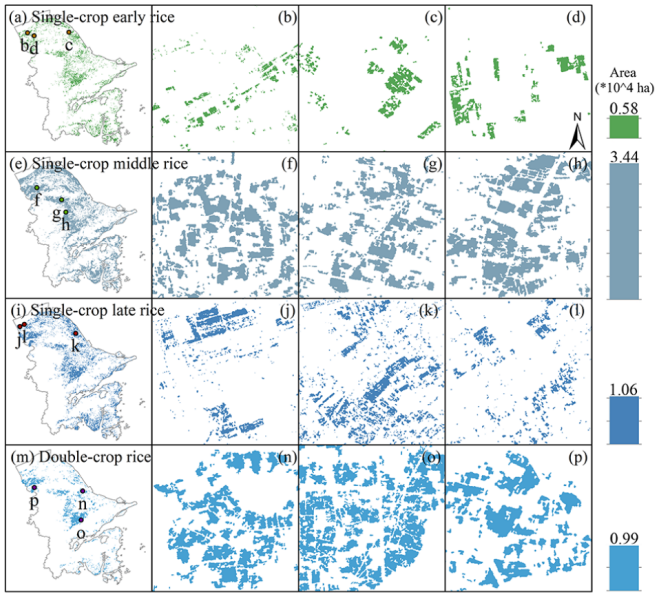


Fig. 7. Single-crop early (a), middle (e), late rice (i), and double-crop rice (m) mapping results in Ningbo City in 2022, the last three columns of figures are zoomed-in sections of the first column figure (b)–(d), (f)–(h), (j)–(l), and (n)–(p).

TABLE III

UA, PA, AND $F1$-SCORES OF EARLY, MIDDLE, AND LATE RICE FROM FOUR EXPERIMENTAL SCENARIOS

		ES 1	ES2	ES 3	ES 4
Single-crop early rice	UA	0.76	0.79	0.81	0.87
	PA	0.75	0.81	0.88	0.84
	F1-score	0.75	0.80	0.84	**0.85**
Single-crop middle rice	UA	0.77	0.84	0.87	0.87
	PA	0.83	0.80	0.81	0.81
	F1-score	0.80	0.82	0.84	**0.84**
Single-crop late rice	UA	0.84	0.89	0.84	0.89
	PA	0.81	0.90	0.88	0.90
	F1-score	0.82	0.89	0.86	**0.89**
Double-crop rice	UA	0.81	0.84	0.87	0.86
	PA	0.91	0.91	0.86	0.93
	F1-score	0.86	0.87	0.86	**0.89**
	Mean F1-score	0.81	0.85	0.85	0.87

types of features were used at the same time (ES4) were the maximum of the four experimental scenarios, which were 0.85, 0.84, 0.89, and 0.89, respectively.

B. Optimal Annual Map of Paddy Rice in 2022

ES4 was determined to be the most favorable experimental scenario based on the mean $F1$-score. Fig. 7 illustrates the distribution of single-crop rice (early, middle, or late rice) and double-crop rice (early and late rice) in Ningbo in 2022 based on ES4. The last three columns of figures are zoomed-in sections of the first column figure, intended to provide readers with an understanding of the small-scale distribution of rice fields (Fig. 7). In that year, the planted areas for single-crop early, middle rice, late rice, and double-crop rice in Ningbo were 5772.37, 34433.81, 10635.04, and 9932.72 ha, respectively (Fig. 7).

TABLE IV

ACCURACIES OF RICE MAPPING AT CASE STUDY AREA IN 2020

Type	Number of samples	UA	PA	F1-score	Mean of F1-score
Single-crop early rice	100	0.79	0.79	0.79	0.84
Single-crop middle rice	100	0.89	0.87	0.88	
Single-crop late rice	100	0.81	0.85	0.83	
Double-crop rice	100	0.84	0.9	0.87	

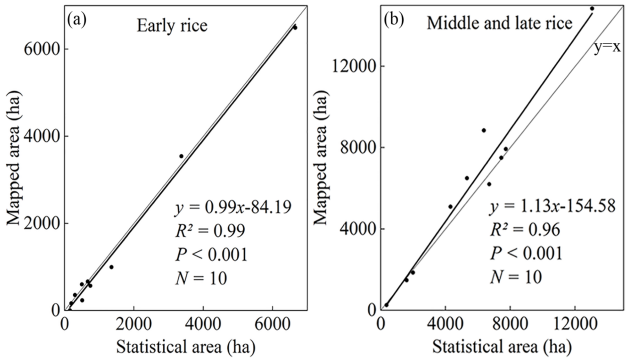


Fig. 8. Comparison of rice area estimates by district and county between the rice map in 2020 and the agricultural statistical data reported for 2020. Since the Ningbo Statistical Yearbook does not make a detailed distinction between middle and late rice, we have combined the areas of middle and late rice for ease of comparison with the statistics. (a) Comparison between the SAR-derived early rice area and the corresponding agricultural statistical data, while (b) compares the sum of the SAR-derived middle and late rice areas with the corresponding agricultural statistical data.

V. DISCUSSION

A. Reliability of Paddy Rice Map

Accuracy assessment demonstrated the high accuracy of the early, middle, and late rice maps. To verify that the rice detection framework proposed in this article can be applied in other years, we supplemented the experiment of rice identification in Ningbo in 2020. The results showed that the $F1$-scores of single-crop early, middle, late rice, and double-crop rice in Ningbo in 2020 were 0.79, 0.88, 0.83, and 0.87, respectively, which proved that the method of this article has strong stability in identifying rice in different years. According to Table IV, the relatively low accuracy of single-crop early rice among the four rice types may be due to the following factors. First, single-crop early rice in the study area was the smallest and dispersed, and it was mostly planted in small fields. This resulted in single-crop early rice corresponding to more mixed image pixels in the Sentienl-1 image, which increased the difficulty of identification. Second, early rice was seeded and transplanted in the spring, which is consistent with the time of grass rejuvenation. In particular, grasslands with standing water in the lowlands may show a pattern of backscattering coefficient change similar to that of early rice, thus affecting the recognition accuracy of early rice.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

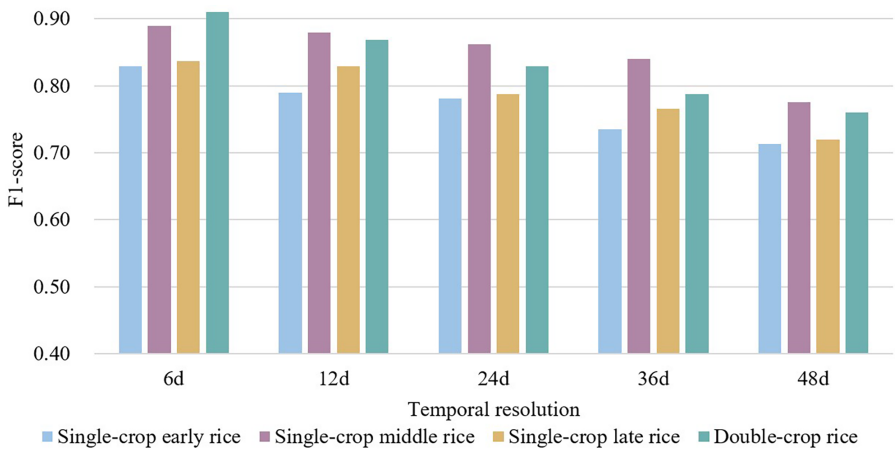


Fig. 9. Variation of precision of early, middle, and late rice with temporal resolution of SAR data.

Fig. 8 shows the comparison between SAR-derived rice area and statistics for each district and county. Since the Ningbo Statistical Yearbook does not make a detailed distinction between middle and late rice, we have combined the areas of middle and late rice for ease of comparison with the statistics. The results of the linear regression analysis are presented in Fig. 8(a) and (b). The correlation coefficient (R^2) for early rice is 0.99, while for middle and late rice, R^2 is 0.96. Both correlations were found to be significant at the 99.9% confidence level. Based on these findings, we can conclude that the results of rice acreage obtained from Sentinle-1 SAR images in this study are generally consistent with the statistical results of the government.

The successful implementation of our study can be attributed to the availability of SAR data, four types of feature images focusing on rice in multiple cropping modes, and a realistic training sample set. The openly obtainable Sentinel-1 SAR data over time function as the basis for precise rice recognition. In contrast to solitary-temporal data, the time-series SAR data can record the phenological data during the complete growth period of rice, thereby allowing precise surveillance of rice in various cropping modes [38].

Second, the four types of features extracted are key to accurately identifying single-crop early, middle, late rice, and double-crop rice. They highlight the specificity of rice relative to other land cover types in terms of four aspects: backscatter coefficient, topography, statistical characteristics, and phenology. Specifically, the difference in backscatter coefficients between rice and nonrice is the basis for rice identification. Topographic features have been shown to contribute to improved accuracy in both early and middle rice. The inclusion of statistical characteristics significantly enhances the accuracy of middle rice. Moreover, the integration of phenological parameters pertaining to rice presents the most optimal means of recognizing early, middle, and late stages of rice growth. It is imperative to acknowledge that the caliber of training samples has a direct impact on the outcomes of the random forest algorithm's training process and the precision of rice classification, underscoring their fundamental role in rice mapping. In order to guarantee trustworthy findings, we amassed a substantial volume of sample data for both single-crop rice and double-crop rice, alongside various other land cover categories. These valuable data were acquired through on-site photographs capturing rice fields, unmanned aerial vehicle (UAV) imagery, and imagery sourced from Google Earth.

B. Sources of Errors and Potential Improvements

The accuracy of rice mapping can be influenced by multiple factors, encompassing the spatial and temporal resolution of time-series SAR data, the methods used for smoothing time-series data, and preprocessing parameters, as well as possible interferences from various land cover categories.

The accuracy in discerning rice is contingent on the temporal resolution of time-series SAR imagery, as it directly impacts their capacity to characterize rice phenology [19]. We sampled Sentinel-1A/B data with a six-day interval in 2020 to generate time-series datasets with temporal resolutions of 6, 12, 24, 36, and 48 days to quantitatively investigate the effect of temporal resolution of SAR images on the rice recognition accuracy (Fig. 9). The findings revealed a notable correlation between temporal resolution and the accuracy of identifying different types of rice, including single-crop early, middle, late rice, and double-crop rice. Specifically, the accuracy of rice identification improved as the temporal resolution increased. Compared with the $F1$-score corresponding to six-day temporal resolution, when the temporal resolution was 24 days, the $F1$-scores of single-crop early, middle, late, and double-crop rice were still located in the range from 0.78 to 0.86, although the $F1$-scores of single-crop early, middle, late, and double-crop rice decreased. This indicates that basically satisfactory rice identification results can also be obtained using SAR data with 24-day temporal resolution. However, when the temporal resolution is less than 24 days, the $F1$-score of most rice types is less than 0.8, and satisfactory rice identification cannot be achieved.

In addition, time-series data smoothing methods and parameters can affect rice identification accuracy. Zeng et al. [42] summarized the commonly used time-series data smoothing methods, which can be divided into four categories: empirical method, filtering method, data transformation, and curve fitting. Using different smoothing methods or setting different parameters will yield different smoothing results, which have important implications for the subsequent rice mapping. Because if the time-series data are not smoothed well, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

noise cannot be suppressed well. If it is oversmoothed, the unique phenological characteristics of rice may not be displayed. Therefore, suitable denoising methods, parameters, and so on should be determined according to the actual rice cultivation and environmental factors in the study area.

Coastal wetlands, along with other land cover types, can impact the accuracy of identifying rice. This is because these wetlands, which contain a significant amount of water, produce a low backscatter signal that closely resembles that of rice paddies during the rice transplanting period. The growing period of wetland vegetation (*Spartina alterniflora*, *Phragmites australis*, *Schoenoplectus triqueter*, and so on) is from June to August, which overlaps with the growing period of middle rice. To improve the accuracy of rice identification, we eliminated wetland vegetation that would be misclassified as middle rice by eliminating it based on its spatial location in tidal flat areas.

VI. Conclusion

The planting structure of rice with multiple cropping modes is complex and difficult to recognize. Therefore, we developed a framework suitable for recognizing rice in multiple cropping modes and successfully applied it to Ningbo City, where rice is grown in complex cropping modes, with single-crop early, middle, late, and double-crop rice. Four types of features are extracted from this framework, among which SAR backscatter coefficients are the basis for SAR-based rice recognition, statistical characteristics help to distinguish rice from nonrice, rice phenology parameters can distinguish rice in different cropping modes, and terrain features also contribute to rice recognition. In addition, the accuracies of different experimental scenarios showed that topographic features contributed most to improving the recognition accuracy of single-crop early and late rice, statistical characteristics improved the recognition accuracy of single-crop early and middle rice, and the incorporation of the rice phenology parameters resulted in the highest recognition accuracies for single-crop early, middle, late rice, and double-crop rice, with $F1$-scores of 0.85, 0.84, 0.89, and 0.89, respectively. This study recognized rice in multiple cropping modes with high accuracy, which provides a reference scheme for rice recognition in other areas with similar cropping modes.

References

[1] S. Kang et al., "Improving agricultural water productivity to ensure food security in China under changing environment: From research to practice," *Agricult. Water Manage.*, vol. 179, pp. 5–17, Jan. 2017, doi: 10.1016/j.agwat.2016.05.007.

[2] H. Li et al., "An approach to high-resolution Rice paddy mapping using time-series Sentinel-1 SAR data in the mun river Basin, Thailand," *Remote Sens.*, vol. 12, no. 23, p. 3959, Dec. 2020, doi: 10.3390/rs12233959.

[3] L. Yang et al., "Semantic segmentation based on temporal features: Learning of temporal–spatial information from time-series SAR images for paddy Rice mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403216, doi: 10.1109/TGRS.2021.3099522.

[4] H. Yang et al., "Integrating multidimensional feature indices and phenological windows for mapping cropping patterns in complex agricultural landscape regions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7448–7459, 2024, doi: 10.1109/JSTARS.2024.3379216.

[5] J. Dong and X. Xiao, "Evolution of regional to global paddy Rice mapping methods: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 119, pp. 214–227, Sep. 2016, doi: 10.1016/j.isprsjprs.2016.05.010.

[6] J. Dong et al., "Mapping paddy Rice planting area in northeastern Asia with Landsat 8 images, phenology-based algorithm and Google Earth engine," *Remote Sens. Environ.*, vol. 185, pp. 142–154, Nov. 2016, doi: 10.1016/j.rse.2016.02.016.

[7] Z. He et al., "Novel harmonic-based scheme for mapping rice-crop intensity at a large scale using time-series Sentinel-1 and ERA5-land datasets," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406523, doi: 10.1109/TGRS.2024.3387559.

[8] T. Chen et al., "Estimation and temporal and spatial pattern of anthropogenic CH4 emissions in China from 2009 to 2019," *China Environ. Sci.*, vol. 42, no. 12, pp. 5549–5560, May 2022. http://www.zghjkx.com.cn/EN/Y2022/V42/I12/5549.

[9] B. Lin et al., "Mitigation of greenhouse gas emissions in China's agricultural sector: Current status and future perspectives," *Chin. J. Eco-Agric.*, vol. 30, no. 4, pp. 500–515, Feb. 2022, doi: 10.12357/cjea.20210843.

[10] S. S. Hasan, L. Zhen, M. G. Miah, T. Ahamed, and A. Samie, "Impact of land use change on ecosystem services: A review," *Environ. Develop.*, vol. 34, Jun. 2020, Art. no. 100527, doi: 10.1016/j.envdev.2020.100527.

[11] N.-T. Son et al., "A phenological object-based approach for Rice crop classification using time-series Sentinel-1 synthetic aperture radar (SAR) data in Taiwan," *Int. J. Remote Sens.*, vol. 42, no. 7, pp. 2722–2739, Apr. 2021, doi: 10.1080/01431161.2020.1862440.

[12] P. Sidike et al., "DPEN: Deep progressively expanded network for mapping heterogeneous agricultural landscape using WorldView-3 satellite imagery," *Remote Sens. Environ.*, vol. 221, pp. 756–772, Feb. 2019, doi: 10.1016/j.rse.2018.11.031.

[13] A. Mateo-Sanchis, M. Piles, J. Muñoz-Marí, J. E. Adsuara, A. Pérez-Suay, and G. Camps-Valls, "Synergistic integration of optical and microwave satellite data for crop yield estimation," *Remote Sens. Environ.*, vol. 234, Dec. 2019, Art. no. 111460, doi: 10.1016/j.rse.2019.111460.

[14] A. M. Rad, D. Ashourloo, H. S. Shahrabi, and H. Nematollahi, "Developing an automatic phenology-based algorithm for Rice detection using Sentinel-2 time-series data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1471–1481, May 2019, doi: 10.1109/JSTARS.2019.2906684.

[15] L. Blickensdörfer, M. Schwieder, D. Pflugmacher, C. Nendel, S. Erasmi, and P. Hostert, "Mapping of crop types and crop sequences with combined time series of Sentinel-1, Sentinel-2 and Landsat 8 data for Germany," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112831, doi: 10.1016/j.rse.2021.112831.

[16] J. Dong et al., "Tracking the dynamics of paddy Rice planting area in 1986–2010 through time series Landsat images and phenology-based algorithms," *Remote Sens. Environ.*, vol. 160, pp. 99–113, Apr. 2015, doi: 10.1016/j.rse.2015.01.004.

[17] Y. He et al., "Examining Rice distribution and cropping intensity in a mixed single- and double-cropping region in South China using all available Sentinel 1/2 images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102351, doi: 10.1016/j.jag.2021.102351.

[18] J. M. Pe na-Barragán, M. K. Ngugi, R. E. Plant, and J. Six, "Object-based crop identification using multiple vegetation indices, textural features and crop phenology," *Remote Sens. Environ.*, vol. 115, no. 6, pp. 1301–1316, Jun. 2011, doi: 10.1016/j.rse.2011.01.009.

[19] L. Wang et al., "An automated extraction of small- and middle-sized Rice fields under complex terrain based on SAR time series: A case study of Chongqing," *Comput. Electron. Agricult.*, vol. 200, Sep. 2022, Art. no. 107232, doi: 10.1016/j.compag.2022.107232.

[20] K. Clauss, M. Ottinger, and C. Kuenzer, "Mapping Rice areas with Sentinel-1 time series and superpixel segmentation," *Int. J. Remote Sens.*, vol. 39, no. 5, pp. 1399–1420, Nov. 2017, doi: 10.1080/01431161.2017.1404162.

[21] X. Zhang, B. Wu, G. E. Ponce-Campos, M. Zhang, S. Chang, and F. Tian, "Mapping up-to-date paddy Rice extent at 10 m resolution in China through the integration of optical and synthetic aperture radar images," *Remote Sens.*, vol. 10, no. 8, p. 1200, Jul. 2018, doi: 10.3390/rs10081200.

[22] P. Zhan, W. Zhu, T. Zhang, X. Cui, and N. Li, "Impacts of sulfate geoengineering on Rice yield in China: Results from a multimodel ensemble," *Earth's Future*, vol. 7, no. 4, pp. 395–410, Apr. 2019, doi: 10.1029/2018ef001094.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.

[23] G. C. Paul, S. Saha, and T. K. Hembram, "Application of phenology-based algorithm and linear regression model for estimating Rice cultivated areas and yield using remote sensing data in Bansloi river basin, Eastern India," *Remote Sens. Appl., Soc. Environ.*, vol. 19, Aug. 2020, Art. no. 100367, doi: 10.1016/j.rsase.2020.100367.

[24] S. Panigrahy, M. Chakraborty, S. A. Sharma, N. Kundu, S. C. Ghose, and M. Pal, "Early estimation of Rice area using temporal ERS-1 synthetic aperture radar data a case study for the Howrah and Hughly districts of West Bengal, India," *Int. J. Remote Sens.*, vol. 18, no. 8, pp. 1827–1833, May 1997, doi: 10.1080/014311697218133.

[25] K. Lasko, K. P. Vadrevu, V. T. Tran, and C. Justice, "Mapping double and single crop paddy Rice with Sentinel-1A at varying spatial scales and polarizations in Hanoi, Vietnam," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 2, pp. 498–512, Feb. 2018, doi: 10.1109/JSTARS.2017.2784784.

[26] H. Bazzi et al., "Mapping paddy Rice using Sentinel-1 SAR time series in Camargue, France," *Remote Sens.*, vol. 11, no. 7, p. 887, Apr. 2019, doi: 10.3390/rs11070887.

[27] S. Xu et al., "A robust index to extract paddy fields in cloudy regions from SAR time series," *Remote Sens. Environ.*, vol. 285, Feb. 2023, Art. no. 113374, doi: 10.1016/j.rse.2022.113374.

[28] N. You and J. Dong, "Examining earliest identifiable timing of crops using all available Sentinel 1/2 imagery and Google Earth Engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 109–123, Mar. 2020, doi: 10.1016/j.isprsjprs.2020.01.001.

[29] D. Cao et al., "Delineating the Rice crop activities in northeast China through regional parametric synthesis using satellite remote sensing time-series data from 2000 to 2015," *J. Integrative Agricult.*, vol. 20, no. 2, pp. 424–437, Feb. 2021, doi: 10.1016/s2095-3119(20)63458-x.

[30] P. Wei et al., "Rice mapping based on Sentinel-1 images using the coupling of prior knowledge and deep semantic segmentation network: A case study in northeast China from 2019 to 2021," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102948, doi: 10.1016/j.jag.2022.102948.

[31] Z. He, S. Li, Y. Wang, L. Dai, and S. Lin, "Monitoring Rice phenology based on backscattering characteristics of multi-temporal RADARSAT-2 datasets," *Remote Sens.*, vol. 10, no. 2, p. 340, Feb. 2018, doi: 10.3390/rs10020340.

[32] T. Ji, G. Li, H. Yang, R. Liu, and T. He, "Comprehensive drought index as an indicator for use in drought monitoring integrating multi-source remote sensing data: A case study covering the Sichuan–Chongqing region," *Int. J. Remote Sens.*, vol. 39, no. 3, pp. 786–809, Feb. 2018, doi: 10.1080/01431161.2017.1392635.

[33] R. Liu et al., "Evaluating effects of medium-resolution optical data availability on phenology-based Rice mapping in China," *Remote Sens.*, vol. 14, no. 13, p. 3134, Jun. 2022, doi: 10.3390/rs14133134.

[34] C. Zhang et al., "Towards automation of in-season crop type mapping using spatiotemporal crop information and remote sensing data," *Agricult. Syst.*, vol. 201, Aug. 2022, Art. no. 103462, doi: 10.1016/j.agsy.2022.103462.

[35] H. Yang, B. Pan, W. Wu, and J. Tai, "Field-based Rice classification in Wuhua county through integration of multi-temporal Sentinel-1A and Landsat-8 OLI data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 69, pp. 226–236, Jul. 2018, doi: 10.1016/j.jag.2018.02.019.

[36] M. Zhang and H. Lin, "Object-based Rice mapping using time-series and phenological data," *Adv. Space Res.*, vol. 63, no. 1, pp. 190–202, Jan. 2019, doi: 10.1016/j.asr.2018.09.018.

[37] J. Wang et al., "Mapping sugarcane plantation dynamics in Guangxi, China, by time series Sentinel-1, Sentinel-2 and Landsat images," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111951, doi: 10.1016/j.rse.2020.111951.

[38] P. Zhan, W. Zhu, and N. Li, "An automated Rice mapping method based on flooding signals in synthetic aperture radar time series," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112112, doi: 10.1016/j.rse.2020.112112.

[39] N. S. Bureau, *Ningbo Statistical Year Book*. Ningbo, China: China Statistics Press, 2021. [Online]. Available: http://tjj.ningbo.gov.cn/col/col1229042824/

[40] R. Gir, L. Jain, and R. Rai, "Speckle reduction of synthetic aperture radar images using median filter and Savitzky–Golay filter," *Int. J. Comput. Appl.*, vol. 113, no. 11, pp. 38–43, Mar. 2015, doi: 10.5120/19874-1877.

[41] S. R. Krishnan and C. S. Seelamantula, "On the selection of optimum Savitzky–Golay filters," *IEEE Trans. Signal Process.*, vol. 61, no. 2, pp. 380–391, Jan. 2013, doi: 10.1109/TSP.2012.2225055.

[42] L. Zeng et al., "A review of vegetation phenological metrics extraction using time-series, multispectral satellite data," *Remote Sens. Environ.*, vol. 237, Oct. 2020, Art. no. 111511, doi: 10.1016/j.rse.2019.111511.

Lihua Wang received the B.S. degree in GIS from Jilin University, Changchun, China, in 2007, and the Ph.D. degree in physical geography from East China Normal University, Shanghai, China, in 2014.

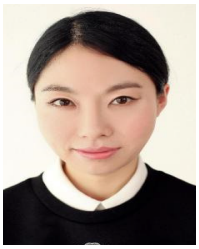
In 2013, she was a Visiting Scholar with European Space Research Institute, European Space Agency, Rome, Italy, under the guidance of renowned Chief Expert Yves-Louis Desnos, focusing on advanced SAR technologies in marine remote sensing. She is currently an Associate Professor with Ningbo University, Ningbo, China. She has published more than 30 journal articles. Her research interests include microwave remote sensing technology of marine and coastal zones.

Tong Wang is currently pursuing the master's degree in surveying and remote sensing engineering with Ningbo University, Ningbo, China.

Her research interests include SAR monitoring of coastal wetlands.

Hao Ma received the M.S. degree in geography from Ningbo University, Ningbo, China, in 2023. He is currently pursuing the Ph.D. degree in estuarine and coastal science with the State Key Laboratory of Estuarine and Coastal Research, East China Normal University, Shanghai, China.

His research interests include coastal wetland biogeomorphic feedbacks, modeling, and prediction of coastal wetland evolution.

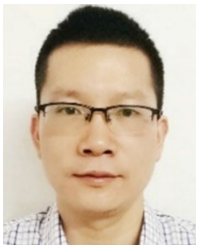


Peng Lu received the B.S. degree in GIS and the Ph.D. degree in geo-information engineering from Jilin University, Changchun, China, in 2007 and 2011, respectively.

She is a Senior Engineer with the College of Earth Exploration Science and Technology, Jilin University. She has published 17 articles in core journals, with ten of them being indexed in SCI/EI. In addition, she has authored one monograph and secured two patents. Her research interests include remote sensing technology and its applications.

Dr. Lu is a member of the Remote Sensing Society of Jilin Province.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.



Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he was with the Department of Applied Mathematics, University of Maryland, College Park, MD, USA, where he was a Visiting Scholar with the famous Prof. John Benedetto to study the dimensionality reduction of hyperspectral image. From 2014 to 2016, he worked with the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, working as a Post-Doctoral Researcher to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he worked with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA, as a Visiting Scholar of hyperspectral image processing. He is currently a Full Professor with Ningbo University, Ningbo, China. He has published more than 80 journal articles. His research interests include hyperspectral image processing with machine learning.

Li Fan received the B.S. degree in land planning from Southwest Agricultural University, Chongqing, China, in 2003, and the M.S. degree in soil science from Southwest University, Chongqing, in 2006.

She is a Senior Engineer with Chongqing Institute of Meteorological Sciences, Chongqing, where she has been engaged in the scientific research and operational work of agro-meteorology and satellite remote sensing since 2006. She has published more than 20 journal articles. Her research interests include remote sensing image classification and information extraction, remote sensing monitoring, and assessment of crop growth process.

Hongmei Wang is currently pursuing the master's degree in geography with Ningbo University, Ningbo, China.

Her research interests include microwave remote sensing technology.

Yuchen Wu is currently pursuing the master's degree in rural development with Ningbo University, Ningbo, China.

Her research interests include agricultural remote sensing.

Yumiao Wang received the B.S. degree in GIS from Anhui Normal University, Wuhu, China, in 2014, and the Ph.D. degree in cartography and GIS from Wuhan University, Wuhan, in 2021.

He is currently an Assistant Researcher with Ningbo University, Ningbo, China. His research interests include agricultural remote sensing and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:36:40 UTC from IEEE Xplore. Restrictions apply.